

Clinical Photo Capture Guide

NEUROINJECT · SPASTICITY INJECTION GUIDE

68 muscles × 4 photos = 272 images

 Patient Position  Probe Placement  Needle Insertion  Ultrasound Image

Save each photo as `<muscleId>-<type>.jpg` (type = `position` · `probe` · `needle` · `us`) into `assets/images/clinical/`. JPG, landscape preferred. The app auto-detects each file and shows an "N/4 captured" counter on the muscle. **61/68** muscles are ultrasound-guided; for the rest, probe/US shots are marked optional.

Pectoralis Major pec-major

Shoulder adduction, internal rotation

- **Patient Position** [pec-major-position.jpg](#)
Position the patient supine with the arm abducted 30-45 degrees and externally rotated slightly to open the chest.
- **Probe Placement** [pec-major-probe.jpg](#)
Patient supine, arm abducted 30°. Place probe transversely on anterior chest, lateral to sternum at 3rd-4th rib. Photo from patient's right side...
- **Needle Insertion** [pec-major-needle.jpg](#)
For the clavicular head: insert the needle 2-3 cm inferior to the lateral clavicle, angled slightly caudally at about 30 degrees to the skin surface.
- **Ultrasound Image** [pec-major-us.jpg](#)
Position the patient supine with the arm slightly abducted. Place the probe transversely on the anterior chest. Depth ≈ 5.5 cm.

Latissimus Dorsi lat-dorsi

Shoulder adduction, internal rotation, extension

- **Patient Position** [lat-dorsi-position.jpg](#)
Position the patient in lateral decubitus with the affected side up, or prone with arm slightly abducted.
- **Probe Placement** [lat-dorsi-probe.jpg](#)
Patient lateral decubitus or prone. Place probe transversely across posterior axillary fold. Photo from behind showing probe on lateral back/axilla.
- **Needle Insertion** [lat-dorsi-needle.jpg](#)
Insert the needle perpendicular to the skin into the mid-belly of the muscle.
- **Ultrasound Image** [lat-dorsi-us.jpg](#)
Identify subcutaneous tissue superficially as a hyperechoic layer.

Biceps Brachii biceps-brachii

Elbow flexion, Forearm supination

- **Patient Position** [biceps-brachii-position.jpg](#)
Position the patient supine or seated with the arm supinated and elbow extended, resting on a support surface.
- **Probe Placement** [biceps-brachii-probe.jpg](#)
Patient supine, arm supinated. Place probe transversely on anterior mid-upper arm. Photo from lateral view showing probe across biceps belly.
- **Needle Insertion** [biceps-brachii-needle.jpg](#)
Insert the needle into the bulk of the muscle belly at the mid-humeral level, perpendicular to the skin surface.
- **Ultrasound Image** [biceps-brachii-us.jpg](#)
Position the patient supine with the arm slightly abducted and externally rotated. Place the probe transversely on the anterior... Depth ≈ 3.5-4.0 cm.

Brachialis brachialis

Pure elbow flexion

- **Patient Position** [brachialis-position.jpg](#)
Position the patient supine or seated with the elbow slightly flexed (about 30 degrees) and the forearm supinated.
- **Probe Placement** [brachialis-probe.jpg](#)
Patient supine, elbow slightly flexed. Place probe transversely on anterior distal arm, 4-6 cm above elbow crease. Photo showing probe on anterior...
- **Needle Insertion** [brachialis-needle.jpg](#)
Insert the needle perpendicular to the skin through the biceps belly, advancing into the brachialis below it.
- **Ultrasound Image** [brachialis-us.jpg](#)
Place the probe transversely across the anterior distal third of the arm, ~4-6 cm proximal to the elbow crease.

Triceps triceps

Elbow extension

- **Patient Position** [triceps-position.jpg](#)
Position the patient seated or prone with the arm resting on a support, elbow flexed to approximately 90 degrees.
- **Probe Placement** [triceps-probe.jpg](#)
Patient seated or prone. Place probe transversely on posterior mid-upper arm. Photo from behind showing probe across triceps belly.
- **Needle Insertion** [triceps-needle.jpg](#)
For the long head: insert the needle into the medial portion of the posterior arm at mid-humeral level, perpendicular to the skin.
- **Ultrasound Image** [triceps-us.jpg](#)
Position the patient prone or seated with the arm supported. Place the probe transversely on the posterior arm. Depth ≈ 3.5-4.5 cm.

Brachioradialis brachioradialis

Elbow flexion (forearm in neutral position)

- **Patient Position** [brachioradialis-position.jpg](#)
Position the patient seated with the forearm resting on a table in neutral position (thumb pointing upward).
- **Probe Placement** [brachioradialis-probe.jpg](#)
Patient seated, forearm neutral on table. Place probe transversely on lateral proximal forearm, 4-6 cm below lateral epicondyle. Photo from lateral...
- **Needle Insertion** [brachioradialis-needle.jpg](#)
Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm, perpendicular to the skin.
- **Ultrasound Image** [brachioradialis-us.jpg](#)
Place the probe transversely on the lateral forearm with the patient's elbow slightly flexed and forearm in neutral (thumb-up)... Depth ≈ 3.5 cm.

Pronator Teres pronator-teres

Forearm pronation

- **Patient Position** [pronator-teres-position.jpg](#)
Position the patient seated with the forearm supinated and the elbow extended, resting on a support surface.
- **Probe Placement** [pronator-teres-probe.jpg](#)
Patient seated, forearm supinated. Place probe transversely on medial proximal forearm, 3-5 cm below medial epicondyle. Photo showing probe on medial...
- **Needle Insertion** [pronator-teres-needle.jpg](#)
Insert the needle 2-3 cm distal to the medial epicondyle, into the bulk of the muscle belly on the medial-to-anterior proximal forearm.
- **Ultrasound Image** [pronator-teres-us.jpg](#)
Position the patient with the elbow extended and forearm supinated to stretch the pronator teres. Depth ≈ 4.5 cm.

Flexor Carpi Radialis fcr

Wrist flexion, Radial deviation

- **Patient Position** [fcr-position.jpg](#)
Position the patient seated with the forearm supinated and resting on a flat surface.
- **Probe Placement** [fcr-probe.jpg](#)
Patient seated, forearm supinated on table. Place probe transversely on central volar forearm at proximal-to-mid third. Photo from above showing...
- **Needle Insertion** [fcr-needle.jpg](#)
Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm, perpendicular to the skin.
- **Ultrasound Image** [fcr-us.jpg](#)
Place the probe transversely on the volar forearm, centered over the palpable FCR tendon. Depth ≈ 2.0 cm.

Flexor Carpi Ulnaris fcu

Wrist flexion, Ulnar deviation

-  **Patient Position** [fcu-position.jpg](#)
Position the patient seated with the forearm supinated or in neutral, resting on a support.
-  **Probe Placement** [fcu-probe.jpg](#)
Patient seated, forearm supinated. Place probe transversely on medial forearm. Photo showing probe on ulnar/medial side of mid-forearm.
-  **Needle Insertion** [fcu-needle.jpg](#)
Insert the needle into the bulk of the muscle belly at the junction of the proximal and middle thirds of the forearm, on the medial (ulnar) aspect.
-  **Ultrasound Image** [fcu-us.jpg](#)
Place the probe transversely on the medial (ulnar) side of the forearm with the forearm fully supinated. Depth $\approx$ 2.5-3.5 cm.

Flexor Digitorum Superficialis fds

Finger flexion at PIP joints, Wrist flexion

-  **Patient Position** [fds-position.jpg](#)
Position the patient seated with the forearm supinated and resting on a flat surface.
-  **Probe Placement** [fds-probe.jpg](#)
Patient seated, forearm supinated on table. Place probe transversely on volar forearm, 5-8 cm below medial epicondyle. Photo showing probe on central...
-  **Needle Insertion** [fds-needle.jpg](#)
Insert the needle at the mid-forearm level, between the FCR and FCU, slightly deeper than the superficial flexor layer.
-  **Ultrasound Image** [fds-us.jpg](#)
Place the probe transversely on the central volar forearm. Depth $\approx$ 2.0-3.0 cm.

Flexor Digitorum Profundus fdp

Finger flexion at DIP joints, Wrist flexion

-  **Patient Position** [fdp-position.jpg](#)
Position the patient seated with the forearm supinated and resting on a flat surface.
-  **Probe Placement** [fdp-probe.jpg](#)
Patient seated, forearm supinated. Place probe transversely on medial-volar mid-forearm. Photo showing probe on deep volar forearm, slightly ulnar.
-  **Needle Insertion** [fdp-needle.jpg](#)
Insert the needle from the medial (ulnar) side of the forearm, just anterior to the palpable subcutaneous ulnar border, at the junction of the...
-  **Ultrasound Image** [fdp-us.jpg](#)
Place the probe transversely on the volar-ulnar forearm at the mid-forearm level. Depth $\approx$ 3.0-4.0 cm.

Flexor Pollicis Longus fpl

Thumb flexion at IP joint

-  **Patient Position** [fpl-position.jpg](#)
Position the patient seated with the forearm supinated and resting flat on a support surface.
-  **Probe Placement** [fpl-probe.jpg](#)
Patient seated, forearm supinated flat. Place probe transversely on mid-volar forearm at junction of middle and distal thirds. Photo showing probe on...
-  **Needle Insertion** [fpl-needle.jpg](#)
Insert the needle at the mid-forearm level, on the radial side, between the brachioradialis (laterally) and the FCR (medially).
-  **Ultrasound Image** [fpl-us.jpg](#)
Place the probe transversely on the volar forearm with the patient supinated. Depth $\approx$ 3.0-3.5 cm.

Adductor Pollicis adductor-pollicis

Thumb adduction (thumb-in-palm deformity)

-  **Patient Position** [adductor-pollicis-position.jpg](#)
Position the patient with the hand resting on a flat surface, palm down (pronated), with the thumb abducted away from the palm.
-  **Probe Placement** [adductor-pollicis-probe.jpg](#)
Patient hand pronated on table. Place small linear or hockey-stick probe transversely across first web space from dorsal approach. Photo from above...
-  **Needle Insertion** [adductor-pollicis-needle.jpg](#)
Insert the needle from the dorsal aspect of the first web space, perpendicular to the plane of the hand.
-  **Ultrasound Image** [adductor-pollicis-us.jpg](#)
Identify the subcutaneous fat and thin dorsal skin as the most superficial layers.

Subscapularis subscapularis

Shoulder internal rotation, Adduction

-  **Patient Position** [subscapularis-position.jpg](#)
Position the patient supine or seated with the arm abducted to 45-60 degrees and externally rotated, resting on a support.
-  **Probe Placement** [subscapularis-probe.jpg](#)
Patient supine, arm abducted 45-60° and externally rotated. Place probe parasagittally in anterior axilla. Photo from front showing probe in axillary...
-  **Needle Insertion** [subscapularis-needle.jpg](#)
Insert the needle through the ANTERIOR axillary fold (formed by pec major), directing it posterolaterally toward the anterior (costal) surface of the...
-  **Ultrasound Image** [subscapularis-us.jpg](#)
Identify the subcutaneous fat and the pectoralis major as the most superficial muscular layer on the anterior chest.

Infraspinatus infraspinatus

 **NOT TYPICALLY INJECTED** — External rotator (reference only)

-  **Patient Position** [infraspinatus-position.jpg](#)
Position the patient seated, leaning forward, or in a prone position with the arms at the sides.
-  **Probe Placement** [infraspinatus-probe.jpg](#)
Patient seated leaning forward or prone. Place probe transversely across infraspinous fossa, parallel to scapular spine. Photo from behind showing...
-  **Needle Insertion** [infraspinatus-needle.jpg](#)
Insert the needle perpendicular to the skin into the center of the infraspinous fossa, targeting the thickest part of the muscle belly.
-  **Ultrasound Image** [infraspinatus-us.jpg](#)
Identify the subcutaneous fat and the thin trapezius muscle as the most superficial layers (the trapezius overlies the...

Teres Major teres-major

Shoulder adduction, Internal rotation, Extension

-  **Patient Position** [teres-major-position.jpg](#)
Position the patient seated or prone with the arm adducted at the side.
-  **Probe Placement** [teres-major-probe.jpg](#)
Patient seated or prone, arm at side. Place probe transversely over posterior axillary fold. Photo from behind showing probe on posterior...
-  **Needle Insertion** [teres-major-needle.jpg](#)
Identify the posterior axillary fold and the inferior angle of the scapula.
-  **Ultrasound Image** [teres-major-us.jpg](#)
Position the patient seated or prone with the arm slightly abducted. Place the probe on the posterior axillary fold. Depth $\approx$ 3.5-4.5 cm.

Pronator Quadratus pronator-quadratus

Forearm pronation (distal)

-  **Patient Position** [pronator-quadratus-position.jpg](#)
Position the patient supine or seated with the forearm supinated and resting on a support, wrist in neutral.
-  **Probe Placement** [pronator-quadratus-probe.jpg](#)
Patient supine, forearm supinated on table. Place probe transversely on volar distal forearm, 3-5 cm proximal to wrist crease. Photo from volar view...
-  **Needle Insertion** [pronator-quadratus-needle.jpg](#)
DORSAL APPROACH (preferred): Insert the needle from the dorsal distal forearm, between the radius and ulna, advancing through the interosseous...
-  **Ultrasound Image** [pronator-quadratus-us.jpg](#)
Place the probe transversely on the volar distal forearm, 3-5 cm proximal to the wrist crease.

Hand Lumbricals hand-lumbricals

MCP flexion + IP extension (intrinsic-plus posture)

-  **Patient Position** [hand-lumbricals-position.jpg](#)
Position the patient with the hand supinated (palm up), fingers relaxed and slightly extended.
-  **Probe Placement** [hand-lumbricals-probe.jpg](#)
Patient palm-up, fingers relaxed. Place hockey-stick probe longitudinally on palm along the metacarpal. Photo from palmar view showing probe on...
-  **Needle Insertion** [hand-lumbricals-needle.jpg](#)
PALMAR APPROACH: Insert the needle into the palm just proximal to the MCP joint of the target finger, on the radial side of the flexor tendon.
-  **Ultrasound Image** [hand-lumbricals-us.jpg](#)
Place the probe longitudinally on the palmar surface along the metacarpal of the target finger.

Quadratus Lumborum quadratus-lumborum

Pelvic hiking, trunk lateral flexion

-  **Patient Position** [quadratus-lumborum-position.jpg](#)
Position the patient prone with a pillow under the abdomen to reduce lumbar lordosis.
-  **Probe Placement** [quadratus-lumborum-probe.jpg](#)
Patient prone, pillow under abdomen. Place curvilinear probe transversely on flank, just above iliac crest. Photo from side showing probe on lateral...
-  **Needle Insertion** [quadratus-lumborum-needle.jpg](#)
Insert the needle perpendicular to the skin, angling slightly medially toward the transverse processes of L3/L4.
-  **Ultrasound Image** [quadratus-lumborum-us.jpg](#)
Identify the subcutaneous fat and thoracolumbar fascia superficially as hyperechoic layers.

Rectus Abdominis rectus-abdominis

Trunk flexion (camptocormia / forward-flexed posture)

-  **Patient Position** [rectus-abdominis-position.jpg](#)
Position the patient supine with the knees slightly bent to relax the abdominal wall.
-  **Probe Placement** [rectus-abdominis-probe.jpg](#)
Patient supine, knees bent. Place linear probe transversely on the anterior abdomen, just lateral to midline at the umbilical level. Photo from the...
-  **Needle Insertion** [rectus-abdominis-needle.jpg](#)
Inject at 2-4 sites distributed along the length of the muscle — typically at the supra-umbilical and infra-umbilical levels bilaterally.
-  **Ultrasound Image** [rectus-abdominis-us.jpg](#)
Place the probe transversely on the anterior abdominal wall, just lateral to the midline at the level of the umbilicus.

Obliques (External & Internal) obliques

Trunk rotation, lateral flexion

-  **Patient Position** [obliques-position.jpg](#)
Position the patient supine or in the lateral decubitus position with the target side up.
-  **Probe Placement** [obliques-probe.jpg](#)
Patient supine or lateral decubitus. Place linear probe transversely on the lateral abdominal wall between the costal margin and iliac crest, along...
-  **Needle Insertion** [obliques-needle.jpg](#)
Target the external oblique and/or internal oblique depending on the clinical pattern — typically inject the internal oblique as it is the primary...
-  **Ultrasound Image** [obliques-us.jpg](#)
Place the probe transversely on the lateral abdominal wall at the level of the umbilicus, along the mid-axillary line.

Paraspinals (Erector Spinae Group) paraspinals

Trunk extension spasticity, axial dystonia, camptocormia treatment

-  **Patient Position** [paraspinals-position.jpg](#)
Position the patient prone with a pillow under the abdomen to reduce lumbar lordosis and open the interlaminar spaces.
-  **Probe Placement** [paraspinals-probe.jpg](#)
Patient prone, pillow under abdomen. Place curvilinear probe parasagittally 2-3 cm lateral to midline spinous processes, or transversely across the...
-  **Needle Insertion** [paraspinals-needle.jpg](#)
Insert the needle perpendicular to the skin or at a slight lateral-to-medial angle, advancing into the muscle belly.
-  **Ultrasound Image** [paraspinals-us.jpg](#)
PARASAGITTAL VIEW: Place the probe longitudinally (parasagittal), 2-3 cm lateral to the midline spinous processes. The transverse...

Gluteus Maximus gluteus-maximus

Hip extension, external rotation

-  **Patient Position** [gluteus-maximus-position.jpg](#)
Position the patient prone or in lateral decubitus with the affected side up.
-  **Probe Placement** [gluteus-maximus-probe.jpg](#)
Patient prone. Place curvilinear probe transversely on upper-outer quadrant of buttock. Photo from behind showing probe position on buttock.
-  **Needle Insertion** [gluteus-maximus-needle.jpg](#)
Insert the needle perpendicular to the skin into the muscle bulk at multiple sites distributed across the belly (3-4 sites for a large muscle).
-  **Ultrasound Image** [gluteus-maximus-us.jpg](#)
Identify the thick subcutaneous fat layer, which can be substantial in this region.

Piriformis piriformis

Hip external rotation, abduction

-  **Patient Position** [piriformis-position.jpg](#)
Position the patient prone with a pillow under the pelvis to slightly flex the hips.
-  **Probe Placement** [piriformis-probe.jpg](#)
Patient prone, pillow under pelvis. Place curvilinear probe transversely along PSIS-to-greater-trochanter line, at midpoint. Photo showing probe...
-  **Needle Insertion** [piriformis-needle.jpg](#)
Insert the needle perpendicular to the skin surface at the midpoint.
-  **Ultrasound Image** [piriformis-us.jpg](#)
Identify the subcutaneous tissue and the thick Gluteus Maximus as a large superficial hypoechoic muscle.

Tensor Fasciae Latae (TFL) tfl

Hip flexion, abduction, internal rotation

-  **Patient Position** [tfl-position.jpg](#)
Position the patient supine or in lateral decubitus with the affected side up.
-  **Probe Placement** [tfl-probe.jpg](#)
Patient supine or lateral. Place probe transversely on anterolateral hip, 2-3 cm below ASIS. Photo showing probe just distal to ASIS on lateral hip.
-  **Needle Insertion** [tfl-needle.jpg](#)
Insert the needle perpendicular to the skin surface directly into the palpated muscle belly.
-  **Ultrasound Image** [tfl-us.jpg](#)
Identify the subcutaneous fat as a superficial hyperechoic layer — it is usually thin in this region.

Rectus Femoris rectus-femoris

Stiff knee gait, Knee extension

-  **Patient Position** [rectus-femoris-position.jpg](#)
Position the patient supine with the knee extended and the thigh relaxed.
-  **Probe Placement** [rectus-femoris-probe.jpg](#)
Patient supine, knee extended. Place probe transversely on anterior mid-thigh. Photo from lateral view showing probe across anterior thigh.
-  **Needle Insertion** [rectus-femoris-needle.jpg](#)
Insert the needle perpendicular to the skin into the bulk of the muscle belly at mid-thigh.
-  **Ultrasound Image** [rectus-femoris-us.jpg](#)
Identify subcutaneous fat and the superficial fascia (fascia lata) as the most superficial hyperechoic layers.

Vastus Medialis vastus-medialis

Knee extension (distal stabilization)

-  **Patient Position** vastus-medialis-position.jpg
Position the patient supine with the knee extended and the thigh relaxed.
-  **Probe Placement** vastus-medialis-probe.jpg
Patient supine, knee extended. Place probe transversely on distal medial thigh, just above patella (VMO area). Photo showing probe on medial...
-  **Needle Insertion** vastus-medialis-needle.jpg
Insert the needle perpendicular to the skin into the bulk of the VMO muscle belly.
-  **Ultrasound Image** vastus-medialis-us.jpg
Identify subcutaneous tissue and the superficial fascia as hyperechoic layers.

Vastus Lateralis vastus-lateralis

Knee extension

-  **Patient Position** vastus-lateralis-position.jpg
Position the patient supine or in lateral decubitus with the affected side up.
-  **Probe Placement** vastus-lateralis-probe.jpg
Patient supine or lateral. Place probe transversely on lateral mid-thigh. Photo from lateral view showing probe on outer thigh.
-  **Needle Insertion** vastus-lateralis-needle.jpg
Insert the needle through the IT band, angling slightly anteriorly to enter the VL muscle belly.
-  **Ultrasound Image** vastus-lateralis-us.jpg
Identify subcutaneous fat and the fascia lata (including the IT band component) as hyperechoic superficial layers.

Vastus Intermedius vastus-intermedius

Knee extension

-  **Patient Position** vastus-intermedius-position.jpg
Position the patient supine with the knee extended and the thigh relaxed.
-  **Probe Placement** vastus-intermedius-probe.jpg
Patient supine, knee extended. Place probe transversely on anterior proximal thigh (same region as rectus femoris but deeper target). Photo from...
-  **Needle Insertion** vastus-intermedius-needle.jpg
Insert the needle perpendicular to the skin through the center of the Rectus Femoris.
-  **Ultrasound Image** vastus-intermedius-us.jpg
Identify subcutaneous fat and fascia lata superficially.

Semimembranosus semimembranosus

Knee flexion

-  **Patient Position** semimembranosus-position.jpg
Position the patient prone with the knee slightly flexed (place a pillow under the ankle).
-  **Probe Placement** semimembranosus-probe.jpg
Patient prone, knee slightly flexed. Place probe transversely on posterior medial thigh at mid-to-distal level. Photo from behind showing probe on...
-  **Needle Insertion** semimembranosus-needle.jpg
Insert the needle from the medial aspect, angling slightly posteriorly and laterally to pass deep to the Semitendinosus.
-  **Ultrasound Image** semimembranosus-us.jpg
Identify subcutaneous tissue and the deep fascia as superficial hyperechoic layers.

Semitendinosus semitendinosus

Knee flexion

-  **Patient Position** semitendinosus-position.jpg
Position the patient prone with the knee slightly flexed (place a pillow under the ankle).
-  **Probe Placement** semitendinosus-probe.jpg
Patient prone, knee slightly flexed. Place probe transversely on posterior mid-thigh. Photo from behind showing probe on posterior thigh, medial side.
-  **Needle Insertion** semitendinosus-needle.jpg
Insert the needle perpendicular to the skin into the mid-belly of the muscle (proximal to the tendinous portion).
-  **Ultrasound Image** semitendinosus-us.jpg
Identify subcutaneous tissue and the deep fascia superficially.

Gastrocnemius (Medial) gastrocnemius-medial

Equinus deformity, Plantarflexion

-  **Patient Position** gastrocnemius-medial-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.
-  **Probe Placement** gastrocnemius-medial-probe.jpg
Patient prone, feet off bed edge. Place probe transversely on proximal medial calf, 2-3 cm below popliteal crease. Photo from behind showing probe on...
-  **Needle Insertion** gastrocnemius-medial-needle.jpg
Insert the needle perpendicular to the skin into the bulk of the medial gastrocnemius belly.
-  **Ultrasound Image** gastrocnemius-medial-us.jpg
Identify subcutaneous fat and the superficial fascia.

Gastrocnemius (Lateral) gastrocnemius-lateral

Equinus deformity, Plantarflexion

-  **Patient Position** gastrocnemius-lateral-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.
-  **Probe Placement** gastrocnemius-lateral-probe.jpg
Patient prone, feet off bed edge. Place probe transversely on proximal lateral calf, 2-3 cm below popliteal crease. Photo from behind showing probe...
-  **Needle Insertion** gastrocnemius-lateral-needle.jpg
Insert the needle perpendicular to the skin into the lateral muscle belly.
-  **Ultrasound Image** gastrocnemius-lateral-us.jpg
Identify subcutaneous fat and superficial fascia.

Soleus (Medial) soleus-medial

Plantarflexion

-  **Patient Position** soleus-medial-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.
-  **Probe Placement** soleus-medial-probe.jpg
Patient prone, feet off edge. Place probe transversely on medial mid-calf, just posterior to tibial border. Photo from behind/medial showing probe on...
-  **Needle Insertion** soleus-medial-needle.jpg
Insert the needle perpendicular to the skin, directed slightly laterally and posteriorly.
-  **Ultrasound Image** soleus-medial-us.jpg
Identify subcutaneous tissue and superficial fascia.

Soleus (Lateral) soleus-lateral

Plantarflexion

-  **Patient Position** soleus-lateral-position.jpg
Position the patient prone with the feet hanging off the edge of the bed.
-  **Probe Placement** soleus-lateral-probe.jpg
Patient prone, feet off edge. Place probe transversely on lateral mid-calf. Photo from behind showing probe on lateral mid-calf.
-  **Needle Insertion** soleus-lateral-needle.jpg
Insert the needle perpendicular to the skin into this lateral soleus bulge at mid-calf level.
-  **Ultrasound Image** soleus-lateral-us.jpg
Identify subcutaneous tissue and superficial fascia.

Flexor Digitorum Longus (FDL) fdl

Claw toes, toe flexion

-  **Patient Position** fdl-position.jpg
Position the patient prone or supine with the leg externally rotated to expose the medial calf.
-  **Probe Placement** fdl-probe.jpg
Patient prone or supine with leg externally rotated. Place probe transversely on medial distal leg, just behind tibial border. Photo showing probe on...
-  **Needle Insertion** fdl-needle.jpg
Insert the needle perpendicular to the skin, directing it toward the posterior aspect of the tibia and fibula.
-  **Ultrasound Image** fdl-us.jpg
Identify subcutaneous tissue and the superficial crural fascia.

Flexor Digitorum Brevis (FDB) fdb

Toe flexion (plantar)

☒ Patient Position fdb-position.jpg

Position the patient prone with the foot hanging off the edge of the bed, or supine with the ankle dorsiflexed to expose the sole.

☐ Probe Placement fdb-probe.jpg

Patient prone, foot hanging off bed. Place probe longitudinally on plantar midfoot from calcaneus toward toes. Photo from plantar view showing probe...

☐ Needle Insertion fdb-needle.jpg

Insert the needle perpendicular to the plantar skin surface.

☐ Ultrasound Image fdb-us.jpg

Identify the thick hyperechoic plantar fascia as the most superficial fibrous layer on the sole.

Adductor Hallucis add-hallucis

Hallux adduction, transverse arch collapse

☒ Patient Position add-hallucis-position.jpg

Position the patient supine with the foot accessible, or prone with the foot dorsiflexed.

☐ Probe Placement add-hallucis-probe.jpg

Patient supine, foot accessible. Place probe longitudinally along 1st-2nd intermetatarsal space on plantar surface. Photo showing probe on plantar...

☐ Needle Insertion add-hallucis-needle.jpg

A dorsal approach is an alternative: insert the needle from the dorsum of the foot between the 1st and 2nd metatarsal shafts, directed plantarward.

☐ Ultrasound Image add-hallucis-us.jpg

Identify the hyperechoic cortex of the 1st and 2nd metatarsal shafts as bright lines with posterior shadowing.

Tibialis Anterior tibialis-anterior

 NOT TYPICALLY INJECTED — Dorsiflexor (reference only)

☒ Patient Position tibialis-anterior-position.jpg

Position the patient supine with the leg extended and the ankle in neutral.

☐ Probe Placement tibialis-anterior-probe.jpg

Patient supine, leg extended. Place probe transversely on proximal anterolateral leg, 2 cm lateral to tibial crest. Photo from front showing probe on...

☐ Needle Insertion tibialis-anterior-needle.jpg

Insert the needle perpendicular to the skin into the largest muscle bulk, approximately 2 cm lateral to the tibial crest.

☐ Ultrasound Image tibialis-anterior-us.jpg

Identify subcutaneous tissue — it is thin on the anterior leg.

Adductor Longus adductor-longus

Hip adduction

☒ Patient Position adductor-longus-position.jpg

Position the patient supine with the hip slightly abducted and externally rotated, and the knee flexed with the sole of the foot resting against the...

☐ Probe Placement adductor-longus-probe.jpg

Patient supine, hip slightly abducted and externally rotated. Place probe transversely on proximal medial thigh, 3-5 cm below pubic tubercle. Photo...

☐ Needle Insertion adductor-longus-needle.jpg

Insert the needle perpendicular to the skin into the muscle belly just lateral to the palpable tendon.

☐ Ultrasound Image adductor-longus-us.jpg

Identify subcutaneous fat and the fascia lata as the superficial layers.

Tibialis Posterior tibialis-posterior

Equinovarus (Plantarflexion + Inversion)

☒ Patient Position tibialis-posterior-position.jpg

Position the patient prone or supine with the leg externally rotated.

☐ Probe Placement tibialis-posterior-probe.jpg

Patient prone or supine with leg externally rotated. Place probe transversely on medial mid-leg, just behind tibia. Photo from medial side showing...

☐ Needle Insertion tibialis-posterior-needle.jpg

Insert the needle directed posterolaterally, sliding along the posterior aspect of the tibia.

☐ Ultrasound Image tibialis-posterior-us.jpg

Identify subcutaneous tissue and the crural fascia superficially.

Adductor Magnus adductor-magnus

Hip adduction, Hip extension

☒ Patient Position adductor-magnus-position.jpg

Position the patient prone with the legs slightly apart, or supine with the hip abducted and externally rotated.

☐ Probe Placement adductor-magnus-probe.jpg

Patient prone, legs slightly apart. Place probe transversely on posteromedial thigh at junction of middle and distal thirds. Photo from behind...

☐ Needle Insertion adductor-magnus-needle.jpg

Insert the needle from the posteromedial aspect of the thigh, angling slightly anteriorly.

☐ Ultrasound Image adductor-magnus-us.jpg

Identify subcutaneous fat and the deep fascia as the superficial layers.

Flexor Hallucis Longus (FHL) fhl

Claw hallux, Equinovarus support

☒ Patient Position fhl-position.jpg

Position the patient prone with the feet hanging off the edge of the bed.

☐ Probe Placement fhl-probe.jpg

Patient prone, feet off edge. Place probe transversely on distal posterolateral calf, adjacent to fibula. Photo from behind showing probe on distal...

☐ Needle Insertion fhl-needle.jpg

Insert the needle lateral to the Achilles tendon, directed toward the posterior fibula.

☐ Ultrasound Image fhl-us.jpg

Identify subcutaneous tissue and the Soleus muscle/aponeurosis superficially.

Biceps Femoris biceps-femoris

Knee flexion, Hip extension

☒ Patient Position biceps-femoris-position.jpg

Position the patient prone with the knee slightly flexed over a pillow or bolster.

☐ Probe Placement biceps-femoris-probe.jpg

Patient prone, knee flexed over pillow. Place probe transversely on lateral posterior thigh at junction of proximal and middle thirds. Photo from...

☐ Needle Insertion biceps-femoris-needle.jpg

Insert the needle perpendicular to the skin into the lateral muscle belly of the posterior thigh.

☐ Ultrasound Image biceps-femoris-us.jpg

Identify subcutaneous fat and the deep fascia (fascia lata) as the superficial layers.

Extensor Hallucis Longus (EHL) ehl

Hallux extension, hitchhiking thumb sign

☒ Patient Position ehl-position.jpg

Position the patient supine with the leg extended and the ankle in neutral.

☐ Probe Placement ehl-probe.jpg

Patient supine, leg extended. Place probe transversely on distal anterolateral leg, ~10 cm above ankle. Photo from front showing probe on anterior...

☐ Needle Insertion ehl-needle.jpg

Insert the needle perpendicular to the skin into the EHL muscle belly.

☐ Ultrasound Image ehl-us.jpg

Identify the thin subcutaneous tissue on the anterior leg.

Iliopsoas iliopsoas

Hip flexion

☐ Patient Position iliopsoas-position.jpg

Position the patient supine with the hip in neutral position and the knee extended.

☐ Probe Placement iliopsoas-probe.jpg

Patient supine, hip neutral. Place curvilinear probe transversely on proximal anterior thigh, 2-3 cm below inguinal ligament, lateral to femoral...

☐ Needle Insertion iliopsoas-needle.jpg

Under ultrasound guidance, insert the needle lateral to the femoral vessels and advance toward the iliopsoas muscle belly.

☐ Ultrasound Image iliopsoas-us.jpg

Identify subcutaneous fat, the fascia lata, and the Sartorius muscle as the most superficial structures laterally.

Adductor Brevis adductor-brevis

Hip adduction (deep, short adductor)

☐ Patient Position adductor-brevis-position.jpg

Position the patient supine with the hip abducted 30-45 degrees, knee flexed, and the leg externally rotated (frog-leg position).

☐ Probe Placement adductor-brevis-probe.jpg

Patient supine, frog-leg position. Place probe transversely on proximal medial thigh at the adductor longus origin. Photo from medial view showing...

☐ Needle Insertion adductor-brevis-needle.jpg

USG is strongly recommended — insert the needle through the adductor longus to reach the adductor brevis layer deep to it.

☐ Ultrasound Image adductor-brevis-us.jpg

Place the probe transversely on the proximal medial thigh with the hip abducted.

Peroneus Longus peroneus-longus

Ankle eversion, Ankle plantarflexion

☐ Patient Position peroneus-longus-position.jpg

Position the patient supine with the leg internally rotated to expose the lateral compartment, or in a lateral decubitus position with the affected...

☐ Probe Placement peroneus-longus-probe.jpg

Patient supine, leg internally rotated. Place probe transversely on lateral proximal leg, 5-8 cm below fibular head. Photo from lateral view showing...

☐ Needle Insertion peroneus-longus-needle.jpg

Insert the needle perpendicular to the skin into the lateral muscle belly, aiming toward the fibula.

☐ Ultrasound Image peroneus-longus-us.jpg

Identify subcutaneous fat and the crural fascia as the superficial layers.

Gracilis gracilis

Hip adduction + knee flexion (crosses both joints)

☐ Patient Position gracilis-position.jpg

Position the patient supine with the hip abducted 30-45 degrees and knee slightly flexed.

☐ Probe Placement gracilis-probe.jpg

Patient supine, frog-leg position. Place probe transversely on proximal medial thigh. Photo from medial view showing probe on inner thigh.

☐ Needle Insertion gracilis-needle.jpg

Insert the needle perpendicular to the skin on the medial thigh into the superficial muscle belly.

☐ Ultrasound Image gracilis-us.jpg

Place the probe transversely on the proximal medial thigh with the hip abducted.

Extensor Digitorum Longus edl

Toe extension, Ankle dorsiflexion

☐ Patient Position edl-position.jpg

Position the patient supine with the leg in neutral position and the foot relaxed.

☐ Probe Placement edl-probe.jpg

Patient supine, leg neutral. Place probe transversely on anterolateral proximal leg at proximal-middle third junction. Photo from anterolateral view...

☐ Needle Insertion edl-needle.jpg

Insert the needle perpendicular to the skin into the anterolateral muscle belly.

☐ Ultrasound Image edl-us.jpg

Identify subcutaneous fat and the crural fascia as the superficial layers.

Foot Lumbricals foot-lumbricals

MTP flexion + IP extension in the foot (contributes to claw toe deformity)

☐ Patient Position foot-lumbricals-position.jpg

Position the patient prone with the foot hanging off the bed, or supine with the ankle dorsiflexed to expose the sole.

☐ Probe Placement foot-lumbricals-probe.jpg

Patient prone or supine with foot exposed. Place hockey-stick probe longitudinally on plantar surface along the metatarsal. Photo from plantar view.

☐ Needle Insertion foot-lumbricals-needle.jpg

Insert the needle from the plantar surface perpendicular to the skin, just proximal to the MTP joint of the target toe, on the medial (tibial) side...

☐ Ultrasound Image foot-lumbricals-us.jpg

Place the probe longitudinally on the plantar surface along the metatarsal of the target toe.

Sternocleidomastoid (SCM) scm

Head rotation to opposite side, ipsilateral lateral flexion

☒ Patient Position scm-position.jpg

Position the patient supine with the head turned slightly to the contralateral side to make the SCM prominent but not maximally stretched.

☐ Probe Placement scm-probe.jpg

Patient supine, head turned slightly contralateral. Place probe transversely on lateral neck at thyroid cartilage level. Photo from front/lateral...

☐ Needle Insertion scm-needle.jpg

Palpate the muscle belly at the mid-cervical level (approximately the level of the thyroid cartilage). Grasp the muscle between thumb and index...

☐ Ultrasound Image scm-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Semispinalis Capitis semispinalis-capitis

Bilateral head and neck extension (retrocollis)

☒ Patient Position semispinalis-capitis-position.jpg

Position the patient seated with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.

☐ Probe Placement semispinalis-capitis-probe.jpg

Patient seated, neck flexed. Place probe transversely on posterior neck at C2-C5 level, 1-2 cm lateral to midline. Photo from behind showing probe on...

☐ Needle Insertion semispinalis-capitis-needle.jpg

This is a DEEP muscle — the needle must traverse the trapezius and splenius capitis to reach it. USG guidance is essential to confirm correct depth...

☐ Ultrasound Image semispinalis-capitis-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Upper Trapezius upper-trapezius

Scapular elevation, head lateral flexion

☒ Patient Position upper-trapezius-position.jpg

Position the patient seated upright with arms relaxed at the sides, or prone with the forehead resting on hands.

☐ Probe Placement upper-trapezius-probe.jpg

Patient seated, arms relaxed. Place probe transversely on upper trapezius ridge, midway between cervical spine and acromion. Photo from behind...

☐ Needle Insertion upper-trapezius-needle.jpg

Insert the needle perpendicular to the skin surface into the midpoint of the palpated muscle bulk, advancing 1-2 cm into the muscle belly.

☐ Ultrasound Image upper-trapezius-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Scalenes (Anterior, Middle, Posterior) scalenes

Ipsilateral lateral neck flexion (laterocollis), accessory inspiration

☒ Patient Position scalenes-position.jpg

Position the patient supine with the head in neutral position or turned slightly to the contralateral side to expose the lateral neck.

☐ Probe Placement scalenes-probe.jpg

Patient supine, head neutral or slightly turned contralateral. Place probe transversely on lateral neck at C5-C7 level, posterior to SCM. Photo from...

☐ Needle Insertion scalenes-needle.jpg

Under ultrasound, identify the target scalene muscle and confirm the location of the brachial plexus, subclavian/vertebral vessels, and phrenic nerve...

☐ Ultrasound Image scalenes-us.jpg

Identify the SCM as the most superficial anterolateral muscle — use it as your anterior landmark.

Levator Scapulae levator-scapulae

Scapular elevation, cervical lateral flexion and rotation

☒ Patient Position levator-scapulae-position.jpg

Position the patient seated upright with arms relaxed, or prone with the forehead resting on hands. Slight contralateral lateral flexion of the neck...

☐ Probe Placement levator-scapulae-probe.jpg

Patient seated, arms relaxed. Place probe transversely over superior angle of scapula, 2-3 cm lateral to C7-T2 spinous processes. Photo from behind...

☐ Needle Insertion levator-scapulae-needle.jpg

Identify the superior angle of the scapula by palpating the superomedial corner of the scapula — this is the most reliable surface landmark for the...

☐ Ultrasound Image levator-scapulae-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Longissimus Capitis longissimus-capitis

Head and neck extension (retrocollis), ipsilateral lateral flexion and rotation

☒ Patient Position longissimus-capitis-position.jpg

Position the patient seated with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.

☐ Probe Placement longissimus-capitis-probe.jpg

Patient seated, neck slightly flexed. Place probe transversely on posterolateral neck at C4-C7 level, 2-4 cm lateral to midline. Photo from behind...

☐ Needle Insertion longissimus-capitis-needle.jpg

This is a DEEP muscle in the erector spinae group — the needle must traverse the trapezius and splenius capitis to reach it. USG guidance is...

☐ Ultrasound Image longissimus-capitis-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Splenius Capitis splenius-capitis

Ipsilateral head rotation and extension

☒ Patient Position splenius-capitis-position.jpg

Position the patient seated upright with the neck slightly flexed to open the posterior cervical space, or prone with the forehead resting on hands.

☐ Probe Placement splenius-capitis-probe.jpg

Patient seated, neck slightly flexed. Place probe transversely on posterolateral neck at C2-C4 level, 2-3 cm lateral to midline. Photo from behind...

☐ Needle Insertion splenius-capitis-needle.jpg

Insert the needle perpendicular to the skin, advancing through the trapezius (1-1.5 cm) and into the splenius capitis beneath it (an additional...

☐ Ultrasound Image splenius-capitis-us.jpg

Identify the skin and subcutaneous fat as the most superficial hyperechoic layer.

Masseter masseter

Jaw clenching, bruxism, oromandibular dystonia (jaw closing type)

- ☐ **Patient Position** [masseter-position.jpg](#)
Position the patient seated upright or supine with the jaw relaxed and slightly open.
- ☐ **Probe Placement** [masseter-probe.jpg](#)
Patient seated upright with jaw relaxed. Place probe transversely on lateral face over the mandibular ramus midway between zygomatic arch and...
- ☐ **Needle Insertion** [masseter-needle.jpg](#)
Ask the patient to clench their teeth forcefully — the masseter will bulge prominently as a thick rectangular mass over the mandibular ramus between...
- ☐ **Ultrasound Image** [masseter-us.jpg](#)
Place the probe transversely on the lateral face over the mandibular ramus, midway between the zygomatic arch and the angle of... Depth ≈ 2.0-3.0 cm.

Temporalis temporalis

Jaw clenching, bruxism, oromandibular dystonia (jaw closing type), temporal headache

- ☐ **Patient Position** [temporalis-position.jpg](#)
Position the patient seated upright or supine with the jaw relaxed.
- ☐ **Probe Placement** [temporalis-probe.jpg](#)
Patient seated upright with jaw relaxed. Place probe transversely in the temporal fossa above the zygomatic arch. Photo from lateral view showing...
- ☐ **Needle Insertion** [temporalis-needle.jpg](#)
Insert the needle perpendicular to the skull surface into the muscle belly at each site. Advance only 0.5-1 cm — the temporalis is thin (5-10 mm)...
- ☐ **Ultrasound Image** [temporalis-us.jpg](#)
Place the probe transversely on the temporal fossa above the zygomatic arch. Apply minimal pressure — the temporalis is thin and... Depth ≈ 1.0-1.5 cm.

Lateral Pterygoid lateral-ptyergoid not typically US-guided

Jaw opening dystonia, jaw protrusion, oromandibular dystonia (jaw opening type), jaw deviation to contralateral side

- ☐ **Patient Position** [lateral-ptyergoid-position.jpg](#)
Position the patient seated upright or supine with the jaw slightly open (to translate the condyle forward and open the coronoid notch).
- ☐ **Probe Placement** [lateral-ptyergoid-probe.jpg](#)
EMG-guided injection — no ultrasound probe placement. Patient seated with jaw slightly open. Needle inserted through coronoid notch anterior to...
- ☐ **Needle Insertion** [lateral-ptyergoid-needle.jpg](#)
Insert a 25-27G, 1.5-2 inch needle through the coronoid notch, directing it medially and slightly anteriorly toward the infratemporal fossa. Advance...
- ☐ **Ultrasound Image** [lateral-ptyergoid-us.jpg](#)
Optional — not typically ultrasound-guided.

Medial Pterygoid medial-ptyergoid

Jaw clenching, oromandibular dystonia (jaw closing type), synergist with masseter for jaw elevation

- ☐ **Patient Position** [medial-ptyergoid-position.jpg](#)
Position the patient seated upright or supine with the jaw slightly open.
- ☐ **Probe Placement** [medial-ptyergoid-probe.jpg](#)
Patient seated with jaw slightly open. For US guidance: place probe in submandibular region below mandibular angle, angled superiorly. For EMG:...
- ☐ **Needle Insertion** [medial-ptyergoid-needle.jpg](#)
SUBMANDIBULAR (EXTRAORAL) APPROACH (preferred for US guidance): Palpate the medial surface of the mandibular angle from the submandibular region....
- ☐ **Ultrasound Image** [medial-ptyergoid-us.jpg](#)
Place the probe in the submandibular region below and medial to the angle of the mandible, angled superiorly. Depth ≈ 2.5-4.0 cm.

Orbicularis Oculi orbicularis-oculi not typically US-guided

Blepharospasm (involuntary forceful eye closure), hemifacial spasm

- ☐ **Patient Position** [orbicularis-oculi-position.jpg](#)
Position the patient seated upright in a well-lit room. The patient should be relaxed with eyes gently closed or in their resting blepharospasm...
- ☐ **Probe Placement** [orbicularis-oculi-probe.jpg](#)
Landmark-guided injection — no ultrasound used. Patient seated upright in well-lit room. Six injection sites distributed around the orbital rim: 3...
- ☐ **Needle Insertion** [orbicularis-oculi-needle.jpg](#)
Insert the needle subcutaneously (bevel up) at each site, directed AWAY from the globe. The muscle is extremely thin (1-2 mm) and lies just beneath...
- ☐ **Ultrasound Image** [orbicularis-oculi-us.jpg](#)
Optional — not typically ultrasound-guided.

Corrugator Supercilii corrugator-supercilii not typically US-guided

Blepharospasm (glabellar component), brow furrowing, glabellar dystonia

- ☐ **Patient Position** [corrugator-supercilii-position.jpg](#)
Position the patient seated upright. Ask them to frown forcefully — the corrugator creates the vertical (11 lines) furrows between the brows.
- ☐ **Probe Placement** [corrugator-supercilii-probe.jpg](#)
Landmark-guided injection — no ultrasound used. Patient seated upright, frowning to activate the corrugator. Injection at the medial eyebrow...
- ☐ **Needle Insertion** [corrugator-supercilii-needle.jpg](#)
Insert a 30G needle at a shallow angle (20-30 degrees from the skin surface) directed laterally along the corrugator muscle belly. Advance...
- ☐ **Ultrasound Image** [corrugator-supercilii-us.jpg](#)
Optional — not typically ultrasound-guided.

Procerus procerus not typically US-guided

Blepharospasm (glabellar component), horizontal nasal root furrows, glabellar dystonia

- ☐ **Patient Position** [procerus-position.jpg](#)
Position the patient seated upright. Ask them to 'scrunch' their nose or pull their brows down toward the nose — the procerus creates horizontal...
- ☐ **Probe Placement** [procerus-probe.jpg](#)
Landmark-guided injection — no ultrasound used. Patient seated upright, actively scrunching the nose to identify the procerus. Single midline...
- ☐ **Needle Insertion** [procerus-needle.jpg](#)
Insert a 30G needle perpendicular to the skin or at a shallow angle into the midline nasal root/glabella junction, advancing only 2-4 mm into the...
- ☐ **Ultrasound Image** [procerus-us.jpg](#)
Optional — not typically ultrasound-guided.

Frontalis frontalis not typically US-guided

Brow elevation dystonia, forehead spasm, involuntary brow raising; NOTE: this is primarily a cosmetic target (horizontal forehead lines) and is NOT a typical spasticity target

- ☐ **Patient Position** [frontalis-position.jpg](#)
Position the patient seated upright. Ask them to raise their eyebrows forcefully — the frontalis will contract, creating horizontal forehead lines.
- ☐ **Probe Placement** [frontalis-probe.jpg](#)
Landmark-guided injection — no ultrasound used. Patient seated upright, raising eyebrows to activate frontalis. Four to five injection sites...
- ☐ **Needle Insertion** [frontalis-needle.jpg](#)
Standard cosmetic approach: inject at 4-5 sites across the mid-forehead, each approximately 1.5-2 cm above the brow and spaced 1.5-2 cm apart.
- ☐ **Ultrasound Image** [frontalis-us.jpg](#)
Optional — not typically ultrasound-guided.

Platysma platysma not typically US-guided

Anterocollis (forward head flexion), neck banding, platysmal dystonia, involuntary neck tightening

 **Patient Position** platysma-position.jpg
Position the patient seated upright or supine. Ask them to grimace or depress the mandible with lips retracted to make the platysmal bands prominent.

 **Probe Placement** platysma-probe.jpg
Landmark-guided injection — platysmal bands are identified visually by asking the patient to grimace. Patient seated upright, grimacing to display...

 **Needle Insertion** platysma-needle.jpg
Insert a 30G needle subcutaneously into each platysmal band, directing it along the length of the band. Advance only 2-3 mm — the muscle is...

 **Ultrasound Image** platysma-us.jpg
Optional — not typically ultrasound-guided.

Mentalis mentalis not typically US-guided

Chin dimpling dystonia, involuntary chin puckering, mental crease deepening, lower lip protrusion

 **Patient Position** mentalis-position.jpg
Position the patient seated upright. Ask them to 'pout' or push the lower lip forward — the mentalis will contract, producing the characteristic...

 **Probe Placement** mentalis-probe.jpg
Landmark-guided injection — no ultrasound used. Patient seated upright, pouting to activate the mentalis and display chin dimpling. Inject at the...

 **Needle Insertion** mentalis-needle.jpg
For unilateral dystonia or dimpling, inject the affected side only. For bilateral chin dimpling, inject both sides.

 **Ultrasound Image** mentalis-us.jpg
Optional — not typically ultrasound-guided.